Complete Kinematic Analysis Report

Auto-Generated System

February 3, 2026

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1 Step 1: Transformation Matrices

The transformation matrices derived from DH parameters:

Link 1: T_1

$$\begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & 1.0L\cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & 0 & 1.0L\sin(\theta_1) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 2: T_2

$$\begin{bmatrix} \cos(\theta_2) & 0 & -\sin(\theta_2) & 2.0L\cos(\theta_2) \\ \sin(\theta_2) & 0 & \cos(\theta_2) & 2.0L\sin(\theta_2) \\ 0 & -1 & 0 & 5.0L \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 3: T_3

$$\begin{bmatrix} \cos(\theta_3 + 1.0\pi) & 0 & \sin(\theta_3 + 1.0\pi) & 0 \\ \sin(\theta_3 + 1.0\pi) & 0 & -\cos(\theta_3 + 1.0\pi) & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 4: T_4

$$\begin{bmatrix} \cos(\theta_4) & 0 & -\sin(\theta_4) & 2.5L\cos(\theta_4) \\ \sin(\theta_4) & 0 & \cos(\theta_4) & 2.5L\sin(\theta_4) \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 5: T_5

$$\begin{bmatrix} \cos(\theta_5) & 0 & \sin(\theta_5) & 2.5L\cos(\theta_5) \\ \sin(\theta_5) & 0 & -\cos(\theta_5) & 2.5L\sin(\theta_5) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 6: T_6

$$\begin{bmatrix} \cos(\theta_6 + 1.0\pi) & 0 & -\sin(\theta_6 + 1.0\pi) & 0 \\ \sin(\theta_6 + 1.0\pi) & 0 & \cos(\theta_6 + 1.0\pi) & 0 \\ 0 & -1 & 0 & 2.5L \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 7: T_7

$$\begin{bmatrix} \cos(\theta_7) & 0 & -\sin(\theta_7) & 2.0L\cos(\theta_7) \\ \sin(\theta_7) & 0 & \cos(\theta_7) & 2.0L\sin(\theta_7) \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2 Step 2: Loop Equation Sets

The loop closure equation $T_1 \dots T_7 = I$ is permuted into 7 sets to isolate variables.

Set 1: $T_1 T_2 T_3 T_4 = T_7^{-1} T_6^{-1} T_5^{-1}$

Set 2: $T_2 T_3 T_4 T_5 = T_1^{-1} T_7^{-1} T_6^{-1}$

Set 3: $T_3 T_4 T_5 T_6 = T_2^{-1} T_1^{-1} T_7^{-1}$

Set 4: $T_4 T_5 T_6 T_7 = T_3^{-1} T_2^{-1} T_1^{-1}$

Set 5: $T_5 T_6 T_7 T_1 = T_4^{-1} T_3^{-1} T_2^{-1}$

Set 6: $T_6 T_7 T_1 T_2 = T_5^{-1} T_4^{-1} T_3^{-1}$

Set 7: $T_7 T_1 T_2 T_3 = T_6^{-1} T_5^{-1} T_4^{-1}$

3 Step 3: Simplified Scalar Equations

The scalar equations are simplified and grouped by permutation set.

Simplified Equations (Set 1)

$$\begin{aligned}
\text{Permutation 1, Element (p˙x)} &: L\left(-\frac{5\sin(\theta_4)\sin(\theta_1+\theta_2)}{2}-\frac{5\sin(\theta_7)}{2}+\cos(\theta_1)\right.\\
&\quad\left.-\frac{5\cos(\theta_3)\cos(\theta_4)\cos(\theta_1+\theta_2)}{2}-\frac{5\cos(\theta_6)\cos(\theta_7)}{2}+2\cos(\theta_1+\theta_2)+2\right)\\
&= 0\\
\text{Permutation 1, Element (o˙y)} &: \sin(\theta_1)\sin(\theta_3)\cos(\theta_2)+\sin(\theta_2)\sin(\theta_3)\cos(\theta_1)+\sin(\theta_5)\sin(\theta_6)\\
&= 0\\
\text{Permutation 1, Element (a˙y)} &: -\sin(\theta_1)\sin(\theta_2)\cos(\theta_4)+\sin(\theta_1)\sin(\theta_4)\cos(\theta_2)\cos(\theta_3)\\
&\quad+\sin(\theta_2)\sin(\theta_4)\cos(\theta_1)\cos(\theta_3)+\cos(\theta_1)\cos(\theta_2)\cos(\theta_4)-\cos(\theta_6)\\
&= 0\\
\text{Permutation 1, Element (p˙y)} &: L\left(\sin(\theta_1)+\frac{5\sin(\theta_4)\cos(\theta_1+\theta_2)}{2}-\frac{5\sin(\theta_6)}{2}-\frac{5\sin(\theta_1+\theta_2)\cos(\theta_3)\cos(\theta_4)}{2}\right.\\
&\quad\left.+2\sin(\theta_1+\theta_2)\right)\\
&= 0\\
\text{Permutation 1, Element (o˙z)} &: \frac{\sin(\theta_5)\sin^2(\theta_6)\sin(\theta_7)}{\cos(\theta_6)}-\frac{\sin(\theta_5)\sin(\theta_7)}{\cos(\theta_6)}+\cos(\theta_3)-\cos(\theta_5)\cos(\theta_7)\\
&= 0\\
\text{Permutation 1, Element (a˙z)} &: -\sin(\theta_3)\sin(\theta_4)-\sin(\theta_6)\sin(\theta_7)\\
&= 0\\
\text{Permutation 1, Element (p˙z)} &: 5L\left(\frac{\sin(\theta_3)\cos(\theta_4)}{2}+\frac{\sin(\theta_7)\cos(\theta_6)}{2}-\frac{\cos(\theta_7)}{2}+1\right)\\
&= 0
\end{aligned}$$

Simplified Equations (Set 2)

$$\begin{aligned}
\text{Permutation 2, Element (a˙x)} &: -\frac{\sin^2(\theta_1)\sin(\theta_7)}{\cos(\theta_1)}-\sin(\theta_2)\sin(\theta_4)\sin(\theta_5)-\sin(\theta_3)\cos(\theta_2)\cos(\theta_5)\\
&\quad-\sin(\theta_5)\cos(\theta_2)\cos(\theta_3)\cos(\theta_4)+\frac{\sin(\theta_7)}{\cos(\theta_1)}\\
&= 0\\
\text{Permutation 2, Element (o˙y)} &: \frac{\sin(\theta_1)\sin(\theta_6)\sin^2(\theta_7)}{\cos(\theta_7)}-\frac{\sin(\theta_1)\sin(\theta_6)}{\cos(\theta_7)}+\sin(\theta_2)\sin(\theta_4)\cos(\theta_3)\\
&\quad-\cos(\theta_1)\cos(\theta_6)+\cos(\theta_2)\cos(\theta_4)\\
&= 0\\
\text{Permutation 2, Element (a˙y)} &: -\sin(\theta_1)\sin(\theta_7)-\sin(\theta_2)\sin(\theta_3)\cos(\theta_5)-\sin(\theta_2)\sin(\theta_5)\cos(\theta_3)\cos(\theta_4)\\
&\quad+\sin(\theta_4)\sin(\theta_5)\cos(\theta_2)\\
&= 0\\
\text{Permutation 2, Element (n˙z)} &: \sin(\theta_3)\cos(\theta_4)\cos(\theta_5)+\sin(\theta_5)\cos(\theta_3)+\frac{\sin^2(\theta_6)\sin(\theta_7)}{\cos(\theta_6)}-\frac{\sin(\theta_7)}{\cos(\theta_6)}\\
&= 0\\
\text{Permutation 2, Element (o˙z)} &: -\sin(\theta_3)\sin(\theta_4)-\sin(\theta_6)\sin(\theta_7)\\
&= 0\\
\text{Permutation 2, Element (a˙z)} &: \sin(\theta_3)\sin(\theta_5)\cos(\theta_4)-\cos(\theta_3)\cos(\theta_5)+\cos(\theta_7)\\
&= 0\\
\text{Permutation 2, Element (p˙z)} &: 5L\left(\frac{\sin(\theta_3)\cos(\theta_4)\cos(\theta_5)}{2}+\frac{\sin(\theta_3)\cos(\theta_4)}{2}+\frac{\sin(\theta_5)\cos(\theta_3)}{2}-\frac{\cos(\theta_7)}{2}+1\right)\\
&= 0
\end{aligned}$$

Simplified Equations (Set 3)

$$\begin{aligned}
\text{Permutation 3, Element (a˙x)} &: -\frac{\sin(\theta_1)\sin^2(\theta_2)}{\cos(\theta_2)} + \frac{\sin(\theta_1)}{\cos(\theta_2)} + \sin(\theta_2)\cos(\theta_1) + \sin(\theta_3)\sin(\theta_5)\sin(\theta_6) \\
&\quad - \sin(\theta_4)\cos(\theta_3)\cos(\theta_6) - \sin(\theta_6)\cos(\theta_3)\cos(\theta_4)\cos(\theta_5) \\
&= 0 \\
\text{Permutation 3, Element (p˙x)} &: L\left(-\frac{5\sqrt{2}\sin(\theta_3)\cos\left(\theta_5+\frac{\pi}{4}\right)}{2} - \frac{5\sqrt{2}\sin\left(\theta_5+\frac{\pi}{4}\right)\cos(\theta_3)\cos(\theta_4)}{2} + \cos(\theta_2)\right. \\
&\quad \left. - \frac{5\cos(\theta_3)\cos(\theta_4)}{2} + 2\cos(\theta_1+\theta_2) + 2\right) \\
&= 0 \\
\text{Permutation 3, Element (n˙y)} &: -\sin(\theta_3)\sin(\theta_4)\sin(\theta_6) + \sin(\theta_3)\cos(\theta_4)\cos(\theta_5)\cos(\theta_6) \\
&\quad + \sin(\theta_5)\cos(\theta_3)\cos(\theta_6) - \sin(\theta_7) \\
&= 0 \\
\text{Permutation 3, Element (o˙y)} &: \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5) + \cos(\theta_7) \\
&= 0 \\
\text{Permutation 3, Element (a˙y)} &: -\sin(\theta_3)\sin(\theta_4)\cos(\theta_6) - \sin(\theta_3)\sin(\theta_6)\cos(\theta_4)\cos(\theta_5) \\
&\quad - \sin(\theta_5)\sin(\theta_6)\cos(\theta_3) \\
&= 0 \\
\text{Permutation 3, Element (p˙y)} &: 5L\left(-\frac{\sqrt{2}\sin(\theta_3)\sin\left(\theta_5+\frac{\pi}{4}\right)\cos(\theta_4)}{2} - \frac{\sin(\theta_3)\cos(\theta_4)}{2}\right. \\
&\quad \left. + \frac{\sqrt{2}\cos(\theta_3)\cos\left(\theta_5+\frac{\pi}{4}\right)}{2} - 1\right) \\
&= 0 \\
\text{Permutation 3, Element (o˙z)} &: -\frac{\sin^2(\theta_1)\sin(\theta_2)\sin(\theta_7)}{\cos(\theta_1)} + \sin(\theta_1)\sin(\theta_7)\cos(\theta_2) \\
&\quad + \frac{\sin(\theta_2)\sin(\theta_7)}{\cos(\theta_1)} - \sin(\theta_4)\sin(\theta_5) \\
&= 0 \\
\text{Permutation 3, Element (a˙z)} &: -\sin(\theta_1)\sin(\theta_2) + \sin(\theta_4)\sin(\theta_6)\cos(\theta_5) + \cos(\theta_1)\cos(\theta_2) - \cos(\theta_4)\cos(\theta_6) \\
&= 0 \\
\text{Permutation 3, Element (p˙z)} &: L\left(-\sin(\theta_2) + \frac{5\sqrt{2}\sin(\theta_4)\sin\left(\theta_5+\frac{\pi}{4}\right)}{2} + \frac{5\sin(\theta_4)}{2} - 2\sin(\theta_1+\theta_2)\right) \\
&= 0
\end{aligned}$$

Simplified Equations (Set 4)

$$\begin{aligned}
\text{Permutation 4, Element (a˙x)} &: -\sin(\theta_3) - \sin(\theta_4)\sin(\theta_6)\sin(\theta_7) - \sin(\theta_5)\cos(\theta_4)\cos(\theta_7) \\
&\quad + \sin(\theta_7)\cos(\theta_4)\cos(\theta_5)\cos(\theta_6) \\
&= 0 \\
\text{Permutation 4, Element (n˙y)} &: -\frac{\sin^2(\theta_1)\sin(\theta_2)}{\cos(\theta_1)} + \sin(\theta_1)\cos(\theta_2) + \frac{\sin(\theta_2)}{\cos(\theta_1)} - \sin(\theta_4)\sin(\theta_5)\sin(\theta_7) \\
&\quad - \sin(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7) - \sin(\theta_6)\cos(\theta_4)\cos(\theta_7) \\
&= 0 \\
\text{Permutation 4, Element (o˙y)} &: \sin(\theta_1)\sin(\theta_2) - \sin(\theta_4)\sin(\theta_6)\cos(\theta_5) - \cos(\theta_1)\cos(\theta_2) + \cos(\theta_4)\cos(\theta_6) \\
&= 0 \\
\text{Permutation 4, Element (a˙y)} &: -\sin(\theta_4)\sin(\theta_5)\cos(\theta_7) + \sin(\theta_4)\sin(\theta_7)\cos(\theta_5)\cos(\theta_6) \\
&\quad + \sin(\theta_6)\sin(\theta_7)\cos(\theta_4) \\
&= 0
\end{aligned}$$

$$\text{Permutation 4, Element (p˙y)} : L\left(-\sin(\theta_2) - 2\sin(\theta_4)\sin(\theta_5)\sin(\theta_7) + \frac{5\sqrt{2}\sin(\theta_4)\sin\left(\theta_5 + \frac{\pi}{4}\right)}{2} - 2\sin(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7) + \frac{5\sin(\theta_4)}{2} - 2\sin(\theta_6)\cos(\theta_4)\cos(\theta_7)\right) = 0$$

$$\text{Permutation 4, Element (o˙z)} : -\frac{\sin(\theta_1)\sin^2(\theta_2)\sin(\theta_3)}{\cos(\theta_2)} + \frac{\sin(\theta_1)\sin(\theta_3)}{\cos(\theta_2)} + \sin(\theta_2)\sin(\theta_3)\cos(\theta_1) + \sin(\theta_5)\sin(\theta_6) = 0$$

$$\text{Permutation 4, Element (a˙z)} : -\sin(\theta_5)\sin(\theta_7)\cos(\theta_6) + \cos(\theta_3) - \cos(\theta_5)\cos(\theta_7) = 0$$

$$\text{Permutation 4, Element (p˙z)} : L\left(-\sin(\theta_3)\cos(\theta_2) - 2\sin(\theta_3) + 2\sin(\theta_5)\cos(\theta_6)\cos(\theta_7) - 2\sin(\theta_7)\cos(\theta_5) - 5\cos(\theta_3) + \frac{5\sqrt{2}\cos\left(\theta_5 + \frac{\pi}{4}\right)}{2}\right) = 0$$

Simplified Equations (Set 5)

$$\text{Permutation 5, Element (a˙x)} : \frac{\sin(\theta_3)\sin^2(\theta_4)}{\cos(\theta_4)} - \frac{\sin(\theta_3)}{\cos(\theta_4)} - \sin(\theta_5)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_5)\cos(\theta_6) = 0$$

$$\text{Permutation 5, Element (n˙y)} : -\sin(\theta_1)\sin(\theta_5)\sin(\theta_6) + \frac{\sin^2(\theta_2)\sin(\theta_3)}{\cos(\theta_2)} - \frac{\sin(\theta_3)}{\cos(\theta_2)} - \sin(\theta_5)\cos(\theta_1)\cos(\theta_6)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_1)\cos(\theta_5) = 0$$

$$\text{Permutation 5, Element (o˙y)} : \sin(\theta_1)\sin(\theta_5)\cos(\theta_6)\cos(\theta_7) - \sin(\theta_1)\sin(\theta_7)\cos(\theta_5) - \sin(\theta_2)\sin(\theta_3) - \sin(\theta_5)\sin(\theta_6)\cos(\theta_1) = 0$$

$$\text{Permutation 5, Element (a˙y)} : \sin(\theta_5)\sin(\theta_7)\cos(\theta_6) - \cos(\theta_3) + \cos(\theta_5)\cos(\theta_7) = 0$$

$$\text{Permutation 5, Element (o˙z)} : \sin(\theta_1)\sin(\theta_6)\cos(\theta_7) + \frac{\sin(\theta_2)\sin^2(\theta_3)\sin(\theta_4)}{\cos(\theta_3)} - \frac{\sin(\theta_2)\sin(\theta_4)}{\cos(\theta_3)} + \cos(\theta_1)\cos(\theta_6) - \cos(\theta_2)\cos(\theta_4) = 0$$

$$\text{Permutation 5, Element (a˙z)} : \sin(\theta_3)\sin(\theta_4) + \sin(\theta_6)\sin(\theta_7) = 0$$

$$\text{Permutation 5, Element (p˙z)} : L\left(\sin(\theta_1)\cos(\theta_6) - 5\sin(\theta_3)\sin(\theta_4) + 2\sin(\theta_4)\cos(\theta_3) - \sin(\theta_6)\cos(\theta_1)\cos(\theta_7) - 2\sin(\theta_6)\cos(\theta_7)\right) = 0$$

Simplified Equations (Set 6)

$$\begin{aligned}
\text{Permutation 6, Element (p˙x)} : L\Bigg(&-\sin(\theta_1)\sin(\theta_6) - 2\sin(\theta_6)\sin(\theta_1+\theta_2) + 5\sin(\theta_7)\cos(\theta_6) \\
&- \cos(\theta_1)\cos(\theta_6)\cos(\theta_7) + \frac{5\cos(\theta_5)}{2} - 2\cos(\theta_6)\cos(\theta_7)\cos(\theta_1+\theta_2) \\
&- 2\cos(\theta_6)\cos(\theta_7) + \frac{5}{2}\Bigg) \\
&= 0 \\
\text{Permutation 6, Element (o˙y)} : &-\sin(\theta_3)\sin(\theta_4) - \sin(\theta_6)\sin(\theta_7) \\
&= 0 \\
\text{Permutation 6, Element (a˙y)} : &-\sin(\theta_1)\sin(\theta_2)\cos(\theta_6) + \sin(\theta_1)\sin(\theta_6)\cos(\theta_2)\cos(\theta_7) \\
&+ \sin(\theta_2)\sin(\theta_6)\cos(\theta_1)\cos(\theta_7) + \cos(\theta_1)\cos(\theta_2)\cos(\theta_6) - \cos(\theta_4) \\
&= 0 \\
\text{Permutation 6, Element (p˙y)} : &L\left(\sin(\theta_1)\cos(\theta_6) + 5\sin(\theta_6)\sin(\theta_7) - \sin(\theta_6)\cos(\theta_1)\cos(\theta_7)\right. \\
&\left.- 2\sin(\theta_6)\cos(\theta_7)\cos(\theta_1+\theta_2) - 2\sin(\theta_6)\cos(\theta_7) + 2\sin(\theta_1+\theta_2)\cos(\theta_6)\right) \\
&= 0 \\
\text{Permutation 6, Element (o˙z)} : &-\frac{\sin(\theta_3)\sin^2(\theta_4)\sin(\theta_5)}{\cos(\theta_4)} + \frac{\sin(\theta_3)\sin(\theta_5)}{\cos(\theta_4)} - \cos(\theta_3)\cos(\theta_5) + \cos(\theta_7) \\
&= 0 \\
\text{Permutation 6, Element (a˙z)} : &\sin(\theta_1)\sin(\theta_7)\cos(\theta_2) + \sin(\theta_2)\sin(\theta_7)\cos(\theta_1) - \sin(\theta_4)\sin(\theta_5) \\
&= 0 \\
\text{Permutation 6, Element (p˙z)} : &L\Bigg(\frac{5\sin(\theta_5)}{2} - \sin(\theta_7)\cos(\theta_1) - 2\sin(\theta_7)\cos(\theta_1+\theta_2) - 2\sin(\theta_7) - 5\cos(\theta_7) \\
&+ \frac{5}{2}\Bigg) \\
&= 0
\end{aligned}$$

Simplified Equations (Set 7)

$$\begin{aligned}
\text{Permutation 7, Element (a˙x)} : &\sin(\theta_1)\sin(\theta_2)\sin(\theta_3)\cos(\theta_7) - \sin(\theta_3)\cos(\theta_1)\cos(\theta_2)\cos(\theta_7) \\
&+ \frac{\sin(\theta_5)\sin^2(\theta_6)}{\cos(\theta_6)} - \frac{\sin(\theta_5)}{\cos(\theta_6)} + \sin(\theta_7)\cos(\theta_3) \\
&= 0 \\
\text{Permutation 7, Element (p˙x)} : &L\Bigg(-5\sin(\theta_7) + \cos(\theta_1)\cos(\theta_7) - \frac{5\cos(\theta_5)\cos(\theta_6)}{2} - \frac{5\cos(\theta_6)}{2} \\
&+ 2\cos(\theta_7)\cos(\theta_1+\theta_2) + 2\cos(\theta_7)\Bigg) \\
&= 0 \\
\text{Permutation 7, Element (n˙y)} : &\sin(\theta_1)\sin(\theta_2)\sin(\theta_7)\cos(\theta_3) + \sin(\theta_3)\cos(\theta_7) - \frac{\sin^2(\theta_4)\sin(\theta_5)}{\cos(\theta_4)} + \frac{\sin(\theta_5)}{\cos(\theta_4)} \\
&- \sin(\theta_7)\cos(\theta_1)\cos(\theta_2)\cos(\theta_3) \\
&= 0 \\
\text{Permutation 7, Element (o˙y)} : &-\sin(\theta_1)\sin(\theta_7)\cos(\theta_2) - \sin(\theta_2)\sin(\theta_7)\cos(\theta_1) + \sin(\theta_4)\sin(\theta_5) \\
&= 0 \\
\text{Permutation 7, Element (a˙y)} : &\sin(\theta_1)\sin(\theta_2)\sin(\theta_3)\sin(\theta_7) - \sin(\theta_3)\sin(\theta_7)\cos(\theta_1)\cos(\theta_2) \\
&- \cos(\theta_3)\cos(\theta_7) + \cos(\theta_5) \\
&= 0 \\
\text{Permutation 7, Element (p˙y)} : &L\Bigg(-\frac{5\sin(\theta_5)}{2} + \sin(\theta_7)\cos(\theta_1) + 2\sin(\theta_7)\cos(\theta_1+\theta_2) + 2\sin(\theta_7) + 5\cos(\theta_7) \\
&- \frac{5}{2}\Bigg) \\
&= 0
\end{aligned}$$

$$\begin{aligned}\text{Permutation 7, Element (o}\dot{\text{z}}\text{)} : &\sin(\theta_1)\sin(\theta_2) + \frac{\sin(\theta_4)\sin^2(\theta_5)\sin(\theta_6)}{\cos(\theta_5)} - \frac{\sin(\theta_4)\sin(\theta_6)}{\cos(\theta_5)} \\ &- \cos(\theta_1)\cos(\theta_2) + \cos(\theta_4)\cos(\theta_6) \\ &= 0\end{aligned}$$

$$\begin{aligned}\text{Permutation 7, Element (a}\dot{\text{z}}\text{)} : &\sin(\theta_1)\sin(\theta_3)\cos(\theta_2) + \sin(\theta_2)\sin(\theta_3)\cos(\theta_1) + \sin(\theta_5)\sin(\theta_6) \\ &= 0\end{aligned}$$

$$\begin{aligned}\text{Permutation 7, Element (p}\dot{\text{z}}\text{)} : &L\left(-\sin(\theta_1) + \frac{5\sin(\theta_6)\cos(\theta_5)}{2} + \frac{5\sin(\theta_6)}{2} - 2\sin(\theta_1+\theta_2)\right) \\ &= 0\end{aligned}$$

4 Step 4: Single Driver Solutions

Attempting to solve sequentially with a single input joint. No complete solution found for single drivers.

5 Step 5: Dual Driver Solutions

Attempting to solve with dual input joints (checking for redundancy).

5.1 Input: θ_3, θ_5

1. **Solve** θ_4 from Permutation 3, Element (p˙y)
   Form: $A\sin(\theta_4) + B\cos(\theta_4) + C = 0$
   Half-angle substitution $t = \tan(\frac{\theta_4}{2}) \Rightarrow (C-B)t^2 + 2At + (B+C) = 0$

$$
\begin{aligned}
&A = 0 \\
&B = 0 \\
&C = 5L\left(\frac{\sqrt{2}\cos\left(\theta_3\right)\cos\left(\theta_5 + \frac{\pi}{4}\right)}{2} - 1\right)
\end{aligned}
$$

2. **Solve** θ_7 from Permutation 2, Element (a˙z)
   Form: $A\sin(\theta_7) + B\cos(\theta_7) + C = 0$
   Half-angle substitution $t = \tan(\frac{\theta_7}{2}) \Rightarrow (C-B)t^2 + 2At + (B+C) = 0$

$$
\begin{aligned}
&A = 0 \\
&B = 1 \\
&C = \sin\left(\theta_3\right)\sin\left(\theta_5\right)\cos\left(\theta_4\right) - \cos\left(\theta_3\right)\cos\left(\theta_5\right)
\end{aligned}
$$

3. **Solve** θ_6 from Permutation 1, Element (n˙z)
   Form: $A\sin(\theta_6) + B\cos(\theta_6) + C = 0$
   Half-angle substitution $t = \tan(\frac{\theta_6}{2}) \Rightarrow (C-B)t^2 + 2At + (B+C) = 0$

$$
\begin{aligned}
&A = 0 \\
&B = 0 \\
&C = \tilde{\infty}\left(\frac{\sin^2\left(\theta_5\right)\sin\left(\theta_7\right)}{\cos\left(\theta_5\right)} - \frac{\sin\left(\theta_7\right)}{\cos\left(\theta_5\right)}\right) + \sin\left(\theta_3\right)\cos\left(\theta_4\right) + \sin\left(\theta_5\right)\cos\left(\theta_7\right)
\end{aligned}
$$

4. **Solve** θ_2 from Permutation 4, Element (p˙x)
   Form: $A\sin(\theta_2) + B\cos(\theta_2) + C = 0$
   Half-angle substitution $t = \tan(\frac{\theta_2}{2}) \Rightarrow (C-B)t^2 + 2At + (B+C) = 0$

$$
\begin{aligned}
&A = 0 \\
&B = 0 \\
&C = L\left(5\sin\left(\theta_3\right) + 2\sin\left(\theta_4\right)\sin\left(\theta_6\right)\cos\left(\theta_7\right) - 2\sin\left(\theta_5\right)\sin\left(\theta_7\right)\cos\left(\theta_4\right) + \frac{5\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos\left(\theta_4\right)}{2} - 2\cos\left(\theta_3\right)\right. \\
&\qquad\qquad\left. - 2\cos\left(\theta_4\right)\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right) + \frac{5\cos\left(\theta_4\right)}{2}\right)
\end{aligned}
$$

5. **Solve** θ_1 from Permutation 1, Element (n˙x)
   Form: $A\sin(\theta_1) + B\cos(\theta_1) + C = 0$
   Half-angle substitution $t = \tan(\frac{\theta_1}{2}) \Rightarrow (C-B)t^2 + 2At + (B+C) = 0$

$$
\begin{aligned}
&A = \sin\left(\theta_2\right)\cos\left(\theta_3\right)\cos\left(\theta_4\right) - \sin\left(\theta_4\right)\cos\left(\theta_2\right) \\
&B = -\sin\left(\theta_2\right)\sin\left(\theta_4\right) - \cos\left(\theta_2\right)\cos\left(\theta_3\right)\cos\left(\theta_4\right) \\
&C = -\frac{\sin^2\left(\theta_5\right)\sin^2\left(\theta_6\right)\sin^2\left(\theta_7\right)}{\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)} + \frac{\sin^2\left(\theta_5\right)\sin^2\left(\theta_6\right)}{\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)} + \frac{\sin^2\left(\theta_5\right)\sin^2\left(\theta_7\right)}{\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)} \\
&\quad - \frac{\sin^2\left(\theta_5\right)}{\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)} + \sin\left(\theta_5\right)\sin\left(\theta_7\right) + \frac{\sin^2\left(\theta_6\right)\sin^2\left(\theta_7\right)}{\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)} \\
&\quad - \frac{\sin^2\left(\theta_6\right)}{\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)} - \frac{\sin^2\left(\theta_7\right)}{\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)} + \frac{1}{\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)}
\end{aligned}
$$

5.2 Input: θ_4, θ_5

1. **Solve** θ_3 from Permutation 3, Element (p˙y)
   Form: $A\sin(\theta_3) + B\cos(\theta_3) + C = 0$
   Half-angle substitution $t = \tan(\frac{\theta_3}{2}) \Rightarrow (C-B)t^2 + 2At + (B+C) = 0$

$$A = 0$$
$$B = 0$$
$$C = -5L$$

2. **Solve** θ_7 from Permutation 2, Element (a˙z)
   Form: $A\sin(\theta_7) + B\cos(\theta_7) + C = 0$
   Half-angle substitution $t = \tan(\frac{\theta_7}{2}) \Rightarrow (C-B)t^2 + 2At + (B+C) = 0$

$$A = 0$$
$$B = 1$$
$$C = \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5)$$

3. **Solve** θ_6 from Permutation 1, Element (n˙z)
   Form: $A\sin(\theta_6) + B\cos(\theta_6) + C = 0$
   Half-angle substitution $t = \tan(\frac{\theta_6}{2}) \Rightarrow (C-B)t^2 + 2At + (B+C) = 0$

$$A = 0$$
$$B = 0$$
$$C = \tilde{\infty}\left(\frac{\sin^2(\theta_5)\sin(\theta_7)}{\cos(\theta_5)} - \frac{\sin(\theta_7)}{\cos(\theta_5)}\right) + \sin(\theta_3)\cos(\theta_4) + \sin(\theta_5)\cos(\theta_7)$$

4. **Solve** θ_2 from Permutation 4, Element (p˙x)
   Form: $A\sin(\theta_2) + B\cos(\theta_2) + C = 0$
   Half-angle substitution $t = \tan(\frac{\theta_2}{2}) \Rightarrow (C-B)t^2 + 2At + (B+C) = 0$

$$A = 0$$
$$B = 0$$
$$C = L\left(5\sin(\theta_3) + 2\sin(\theta_4)\sin(\theta_6)\cos(\theta_7) - 2\sin(\theta_5)\sin(\theta_7)\cos(\theta_4) + \frac{5\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos(\theta_4)}{2} - 2\cos(\theta_3) - 2\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7) + \frac{5\cos(\theta_4)}{2}\right)$$

5. **Solve** θ_1 from Permutation 1, Element (n˙x)
   Form: $A\sin(\theta_1) + B\cos(\theta_1) + C = 0$
   Half-angle substitution $t = \tan(\frac{\theta_1}{2}) \Rightarrow (C-B)t^2 + 2At + (B+C) = 0$

$$A = \sin(\theta_2)\cos(\theta_3)\cos(\theta_4) - \sin(\theta_4)\cos(\theta_2)$$
$$B = -\sin(\theta_2)\sin(\theta_4) - \cos(\theta_2)\cos(\theta_3)\cos(\theta_4)$$
$$C = -\frac{\sin^2(\theta_5)\sin^2(\theta_6)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{\sin^2(\theta_5)\sin^2(\theta_6)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{\sin^2(\theta_5)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_5)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \sin(\theta_5)\sin(\theta_7) + \frac{\sin^2(\theta_6)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_6)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{1}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)}$$

6 Numerical Verification

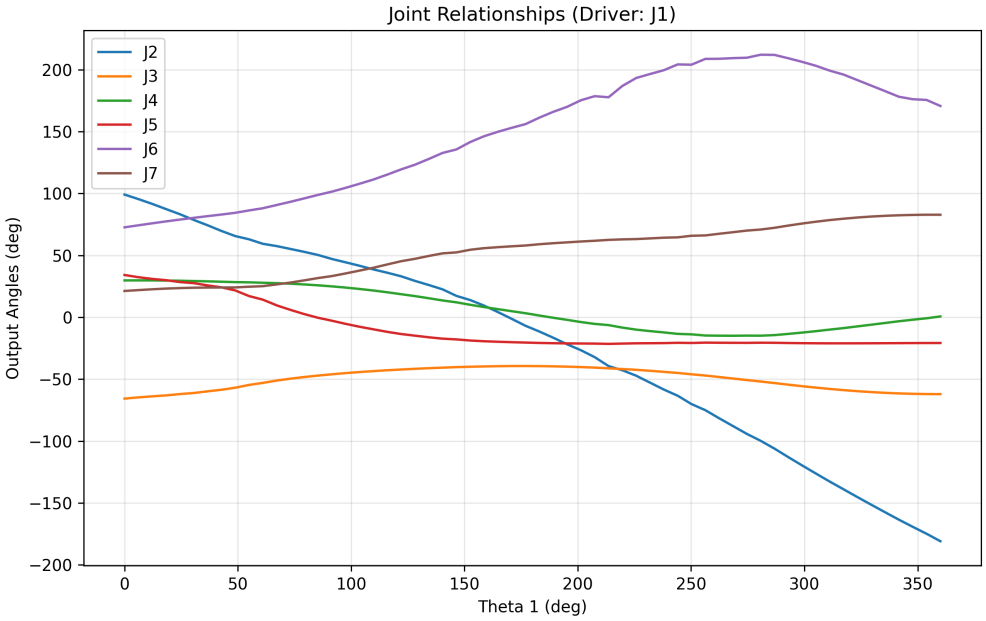


Figure 1: Joint Trajectories